The Full-Dimensional Linear Cahn–Hilliard Spinodal Atlas

Route-A source-local certificate HCS-C304

3 September 2026

Abstract

For every finite $d \geq 1$, $\kappa > 0$, and $\alpha \in \mathbb{R}$, we classify the linear Cahn–Hilliard flow $u_t = -\kappa\Delta^2 u - \alpha\Delta u$ on the 2π-periodic torus. The result gives the self-adjoint analytic trace-class semigroup, exact lattice-shell spectrum, and the complete stable/critical/spinodal atlas.

1 One theorem for the entire parameter space

Let $\mathbb{T}^d = (\mathbb{R}/2\pi\mathbb{Z})^d$, $H = L^2_0(\mathbb{T}^d)$, and

$$A = -\kappa\Delta^2 - \alpha\Delta, \qquad D(A) = H^4(\mathbb{T}^d) \cap H, \quad \kappa > 0, \quad \alpha \in \mathbb{R}.$$

Write $r_d(n) = \#\{k \in \mathbb{Z}^d : |k|^2 = n\}$.

Theorem 1 (Full linear spinodal atlas)**.** *For every finite $d \geq 1$ the operator A is self-adjoint, bounded above, and has compact resolvent. It generates a self-adjoint analytic semigroup $S(t)$, trace class for every $t > 0$. On shell $n > 0$,*

$$Ae_k = \sigma_n e_k, \qquad \sigma_n = \alpha n - \kappa n^2, \tag{1}$$

with multiplicity $r_d(n)$. The mean-zero equilibrium is exponentially stable iff $\alpha < \kappa$, critical iff $\alpha = \kappa$, and spinodally unstable iff $\alpha > \kappa$. Its unstable Morse index and kernel dimension are

$$M = \sum_{n<\alpha/\kappa} r_d(n), \qquad K = \sum_{n=\alpha/\kappa} r_d(n), \tag{2}$$

where absent shells contribute zero. At criticality $K = r_d(1) = 2d$.

The spectral bound is attained precisely on

$$\mathcal{N}_* = \operatorname*{argmax}_{n\geq 1,\ r_d(n)>0} (\alpha n - \kappa n^2), \tag{3}$$

including every tie. For $u_0 \neq 0$, let $\lambda_* = \max\{\sigma_n : P_n u_0 \neq 0\}$. Then

$$e^{-\lambda_* t} S(t) u_0 \longrightarrow \sum_{\sigma_n = \lambda_*} P_n u_0 \quad \text{in } L^2. \tag{4}$$

Every recurrent state is stationary; hence there is no nonstationary periodic solution.

Proof. For $e_k = (2\pi)^{-d/2} e^{ik\cdot x}$, $\Delta e_k = -|k|^2 e_k$, so (1) follows. The real diagonal values tend to $-\infty$ as $|k| \to \infty$, proving self-adjointness, compact resolvent, and an upper bound. The spectral theorem yields the analytic semigroup, while

$$\sum_{k\neq 0} e^{t(\alpha|k|^2 - \kappa|k|^4)} < \infty \qquad (t > 0)$$

by quartic domination, proving trace class. The sign factorization $\sigma_n = n(\alpha - \kappa n)$ proves the chamber and index formulas.

The maximization in (3) is over the whole represented spectrum, not a numerical cutoff. Indeed

$$\sigma_n = \frac{\alpha^2}{4\kappa} - \kappa\left(n - \frac{\alpha}{2\kappa}\right)^2. \tag{5}$$

If $\alpha/\kappa \leq 1$, then for $n > 1$, $\sigma_n - \sigma_1 = (n-1)[\alpha - \kappa(n+1)] < 0$. If $\alpha/\kappa > 1$, shell one has positive rate whereas every $n \geq \alpha/\kappa$ has nonpositive rate. Thus an explicit finite set of represented integers contains every maximizer, and all ties are retained. For a fixed datum the same decay at infinity gives a largest rate in its actual support. Factoring $e^{\lambda_* t}$ and dominated convergence prove (4). Finally, recurrence along $t_j \to \infty$ forces $e^{t_j\sigma_n}\widehat{u}_0(k) \to \widehat{u}_0(k)$ coefficientwise; every nonzero coefficient must have $\sigma_n = 0$, so the state is stationary. □

Source lineage

Cahn and Hilliard introduced the nonuniform free-energy model, and Cahn's spinodal paper is the historical owner token for the instability language. Elliott and Zheng provide classical analytical context. These references establish lineage only.

References

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[2] J. W. Cahn, "On spinodal decomposition," *Acta Metallurgica* 9 (1961), 795–801. DOI: 10.1016/0001-6160(61)90182-1.

[3] C. M. Elliott and S. Zheng, "On the Cahn–Hilliard equation," *Arch. Rational Mech. Anal.* 96 (1986), 339–357. DOI: 10.1007/BF00251803.